

With the use of modern design concepts and readymade library like bootstrap, we can make the web design impressive while saving a lot of time.



Tech Flash

- **Bootstrap** : An HTML and CSS Framework that provides ready-made components and classes to use on a website. It helps easily create responsive websites.
- **Carousel** : A collection of images which appear one after another in a horizontal direction, either automatically or at the click of a button.
- **Cards** : A small rectangular or square shaped container with images or buttons within it.
- **Flexbox** : A tool that helps us create layouts on a page, either by arranging items in a horizontal or vertical direction. Flexbox is mainly used for image galleries, card layouts, etc.

Exercise

Fill in the blanks

- 1 Websites that display fixed content are known as _____.
- 2 The _____ CSS property displays a border at the bottom of an HTML element.
- 3 To create a hover effect, the _____ selector is used.
- 4 The Bootstrap class _____ creates a button with a red background.
- 5 To use Flexbox, we set the display property to _____.
- 6 A Bootstrap carousel displays _____ that play one after another.
- 7 Using _____, we can align HTML components horizontally or vertically.

Activity

9 Design a button

- We will use Bootstrap to design a button. Bootstrap offers a pre-defined "btn btn-danger" class, which automatically applies a pre-defined style to buttons. Let's add this class to all the buttons.
- Open **index.html** and find the **button** tag inside the card element. Within the button tag, add the **btn btn-danger** class.

```
<div class="product_container">
  <h2> Laptops</h2>
  <p>
    We deal with all major Laptop Brands <br>
    <strong> HP, DELL, ASUS, LENOVO </strong>
  </p>
  <button class="btn btn-danger"> Know More</button>
</div>
```

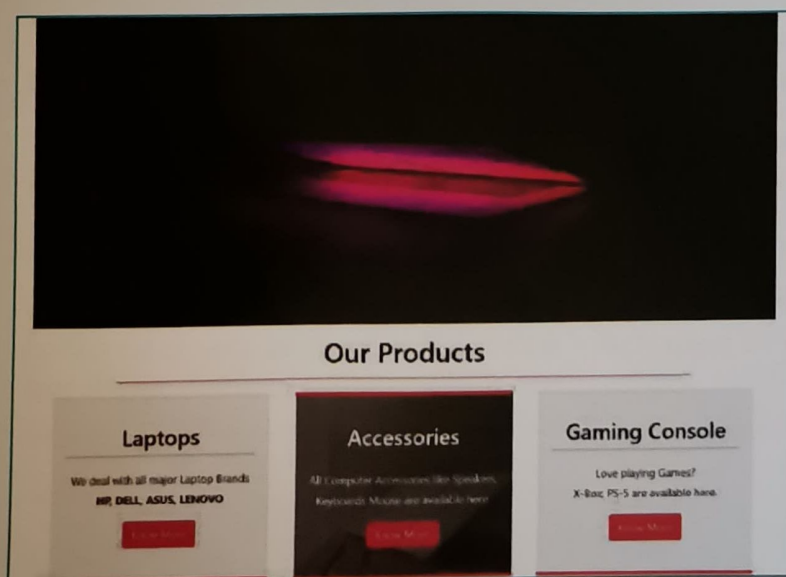


- Similarly, add the class to the other two buttons as well.

Note:

Button design is not linked to the hover effect.

Final Output

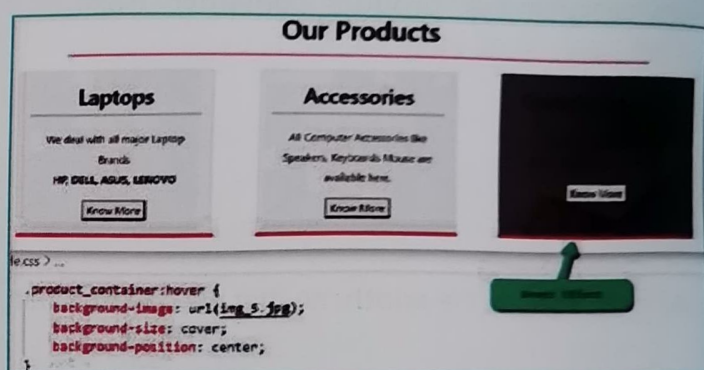


Activity

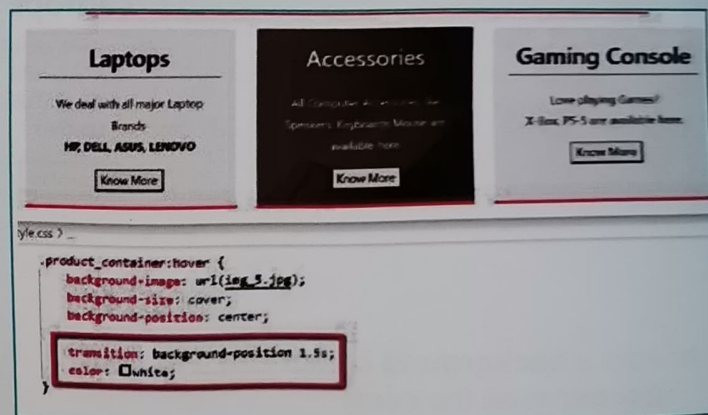
8 Add a hover effect to the cards

The **hover** effect is displayed when we place the cursor over an HTML element. To apply this effect, we use **:hover selector**. In this website, we want the hover effect to play whenever the cursor is placed over any card element.

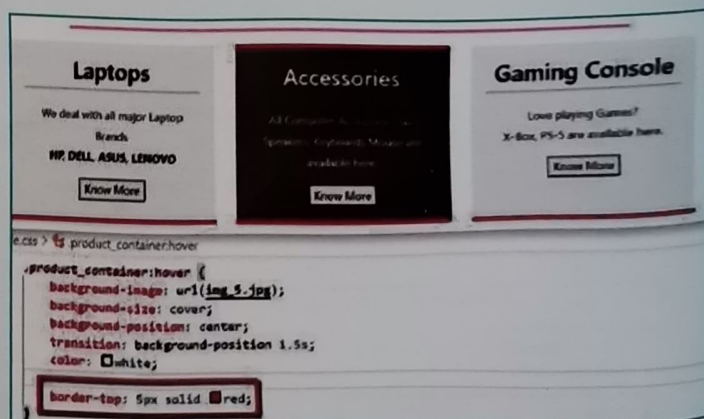
- a. In the style tag, use the **:hover** selector on the **.product_container** class. For now, we will make a background image appear once the cursor is placed over any of the card elements. Use the **background-size** property so that the image occupies the entire card element and the **background-position** property to place the image at the centre.



- b. Right now, the text colour is not in contrast with the background image on hover. Change this using the **color** property. Also, use the **transition** property to make the image slowly appear. This property needs 2 values, first a **CSS property** and secondly the time, which specifies the duration of the effect.



- c. Finally, make a top border to appear on hover.



Activity

7 Design card contents

Each one of these cards will have the same overall design, so we will give the card containers the same class name “**product_container**”. Now, when we specify any design in the “**product_container**” class, it will apply to all three cards.

```
<div class="product_container">
  <h3> Laptops</h3>
  <p>
    We deal with all major Laptop Brands <br>
    <strong>HP, DELL, ASUS, LENOVO </strong>
  </p>
  <button> Know More</button>
</div>

<div class="product_container">
  <h3> Accessories</h3>
  <p>
    All Computer Accessories like Speakers, Keyboards, Mouse are available
  </p>
  <button> Know More</button>
</div>

<div class="product_container">
  <h3> Gaming Consoles</h3>
  <p>
    Love playing Games? <br> X-Box, PS-5 are available here. </p>
  <button> Know More</button>
</div>
```

a. First, specify the **width**, so that we can maintain a fixed size for the cards. Then add **padding**.

```
.product_container {
  width: 350px;
  padding: 20px 40px;
}
```

Laptops

We deal with all major Laptop Brands
HP, DELL, ASUS, LENOVO

Know More

Accessories

All Computer Accessories like Speakers, Keyboards Mouse are available here.

Know More

Gaming Console

Love playing Games?
X-Box, PS-5 are available here.

Know More

b. Add a **background colour** and a **bottom border** to all the cards.

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Accessories

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```
.product_container {
  width: 380px;
  padding: 20px;
  background-color: whitesmoke;
  border-bottom: 5px solid red;
}
```

c. Add a **border** above the card paragraphs and increase the space between paragraph lines using the **line-height** property. Apply these properties to the paragraph tag in the **product_container** div element.

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Accessories

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Gaming Console

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```
.product_container p {
  padding-top: 20px;
  border-top: 1px solid grey;
  line-height: 30px;
}
```


Activity

Using Flex

f. Inside the `<style>` tag, specify the `display: flex` property in the “products” div element.

```
.products {  
  display: flex;  
}
```

Our Products

Laptops

We deal with all major Laptop Brands

HP, DELL, ASUS, LENOVO

[Know More](#)

Accessories

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After we set the display property to flex, the card elements are horizontally aligned.

g. Use the `justify-content: space-evenly` property to arrange the card elements within the “products” container

```
.products {  
  display: flex;  
  justify-content: space-evenly;  
}
```

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Accessories

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h. Add padding to create some space around the cards. Use text-align to display content at the centre of the card.

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Laptops

We deal with all major Laptop Brands

HP, DELL, ASUS, LENOVO

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Accessories

All Computer Accessories like Speakers, Keyboards, Mouse are available here.

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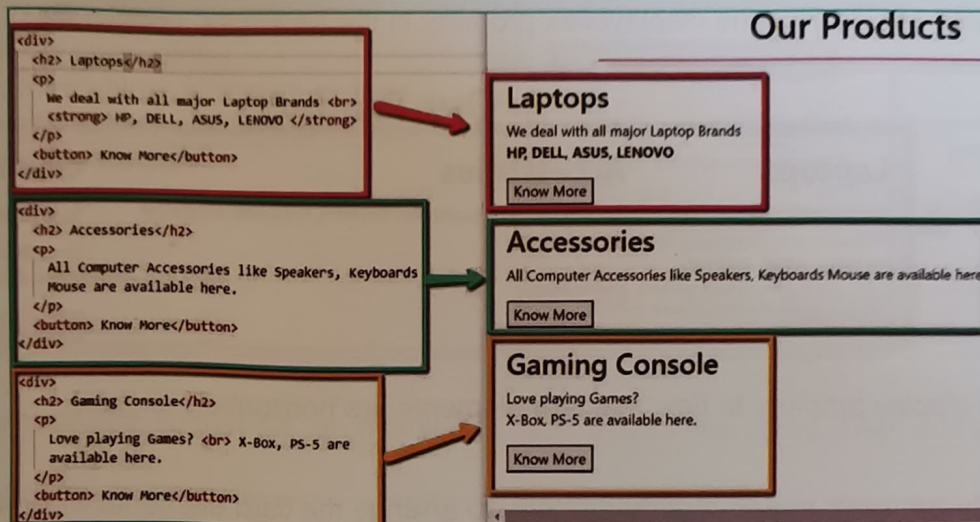
Gaming Console

Love playing Games?
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Activity

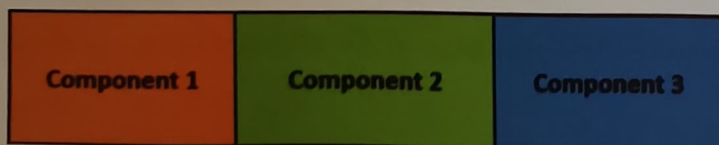
d. Similarly, add content to the remaining cards.



e. Notice that the cards appear one on top of the other. We want them to appear side-by-side. Let's use **Flexbox** for this.

Flexbox

Flexbox, or simply flex, offers an easy way to align HTML components within a webpage. They are aligned in a row-wise (horizontal) or in a column wise (vertical) manner. To convert an HTML container into a Flexbox, simply set the 'display' property to 'flex'.



Horizontal Arrangement



Vertical Arrangement

In our application, all the card elements are inside the div element with the class "products". This makes the "products" div element the parent element. The card elements inside are its child elements.

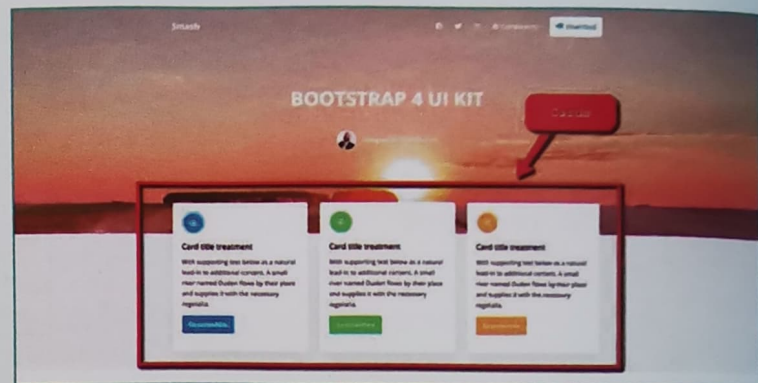
Activity

6 Design a Products Section

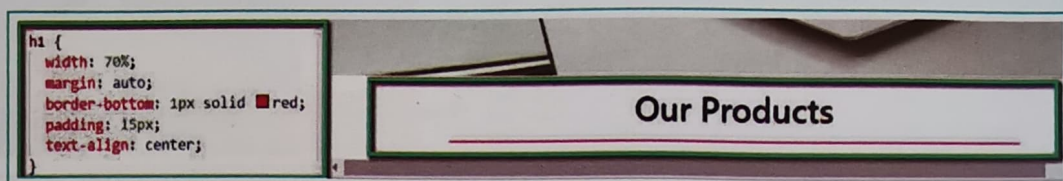
Assuming our computer store deals with laptops, accessories and gaming consoles, we need to display these three product categories on the website. There are different ways to do it, but we will use **Cards**.

Implement Card UI Design with Hover Effect

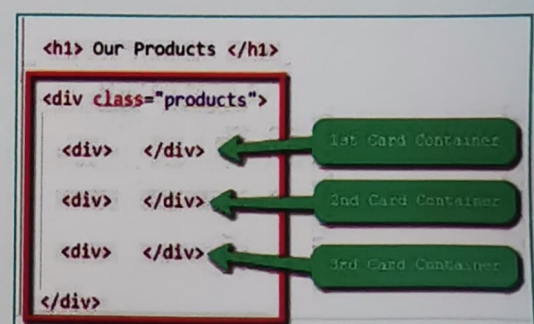
A card is a small rectangular module with images and/or text. It is an entry point for users to learn more about a company's products or services. Cards make the website interface look neat, spacious, and well organised.



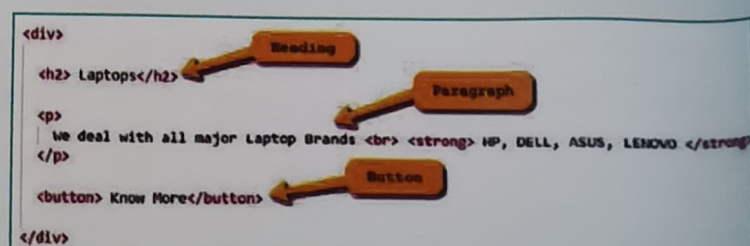
- Add a title for the products section using the `<h1>` tag. For the design, we will add a few CSS properties as per the image below:



- Create a div element with class name "products". This container will house our cards. Inside it, create three more div elements. Each one of these will contain a single card.



- Let's populate the cards with some content. Add a heading, paragraph and a button for each card.



Activity

5 Add CSS design properties to the image carousel

CSS is a language used to design web pages. In this activity, we will use the Internal CSS method to write a CSS Code.

- a. Inside the **<head>** tag, create a **<style>** tag. This HTML tag allows us to write CSS code in the HTML page itself.

```
<html>
  <head>
    <title> E-Commerce Website </title>
    <link rel="stylesheet" href="https://cdn.jsdelivr.net/npm/bootstrap@5.1.3/dist/css/bootstrap.min.css">
    <style>
    </style>
  </head>
  <body>
```

- b. First, set a **width** and **height** to the **carousel_container** div and the images appearing in the carousel.

```
<link rel="stylesheet" href="https://cdn.jsdelivr.net/npm/bootstrap@5.1.3/dist/css/bootstrap.min.css">
<style>
  .carousel_container {
    width: 1100px;
    height: 500px;
  }
  .carousel_container img {
    width: 100%;
    height: 100%;
  }
</style>
```



Setting width and height of the container

Images will cover the entire dimensions of its container

- c. Notice that the container is aligned to the left side of the browser. Use the **margin:auto** property to display it in the centre.

```
<style>
  .carousel_container {
    width: 1100px;
    height: 500px;
    margin: auto;
  }
  .carousel_container img {
    width: 100%;
    height: 100%;
  }
</style>
```



Now we have a carousel with three images, placed at the centre of the browser window. We can already agree that Bootstrap is quite simple and saves a lot of time by providing various ready-made components.

Activity

- d. In the index.html file, create another div element with class name **"carousel_container"**. This div container will house our Bootstrap carousel. Paste the copied code inside it.

```

<div class="main">
  <div class="carousel_container">
    <div id="carouselExampleAutoplaying" class="carousel slide" data-bs-ride="carousel">
      <div class="carousel-inner">
        <div class="carousel-item active">
          
        </div>
        <div class="carousel-item">
          
        </div>
        <div class="carousel-item">
          
        </div>
      </div>
      <button class="carousel-control-prev" type="button" data-bs-target="#carouselExampleAutoplaying" data-bs-slide="prev">
        <span class="carousel-control-prev-icon" aria-hidden="true"></span>
        <span class="visually-hidden">Previous</span>
      </button>
      <button class="carousel-control-next" type="button" data-bs-target="#carouselExampleAutoplaying" data-bs-slide="next">
        <span class="carousel-control-next-icon" aria-hidden="true"></span>
        <span class="visually-hidden">Next</span>
      </button>
    </div>
  </div>
</div>
  
```

4 Adding images to the carousel

We have the code to create a carousel, but we haven't added any images to it yet.

- a. Notice the **** tag with blank **src** attribute within the carousel code. This is where we add images. In HTML, use the **** tag to insert images. Add the image name within the 'src' attribute.

Note:

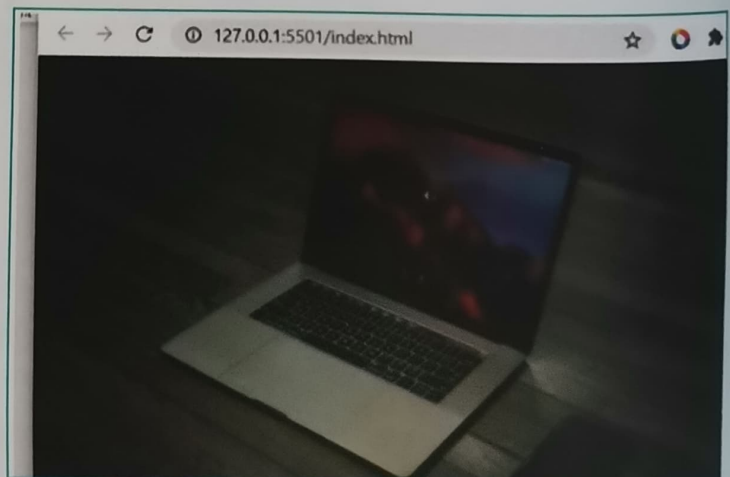
Add any three computer related images into the carousel.

```

<div id="carouselExampleAutoplaying" class="carousel slide" data-bs-ride="carousel">
  <div class="carousel-inner">
    <div class="carousel-item active">
      
    </div>
    <div class="carousel-item">
      
    </div>
    <div class="carousel-item">
      
    </div>
  </div>
  
```

- a. Run the file on the browser. The three images will appear one after the other.

Our carousel works fine at this point. But we can observe in the output that the images cover the entire browser. Let's add a specific height and width to it so the images will look proportionate on the website.



Activity

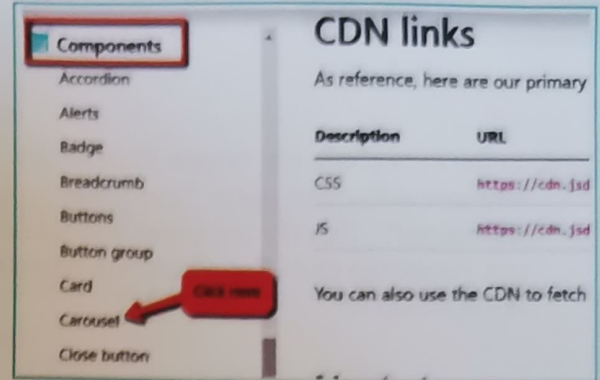
- a. To add the Carousel, move your mouse over the sidebar and scroll down to the Components section and click on Carousel.

Note:

Once you are on the Carousel window, you will find instructions for displaying different types of Carousel. To keep things simple, we will choose an auto playing Carousel. But feel free to test out the other types of carousels.

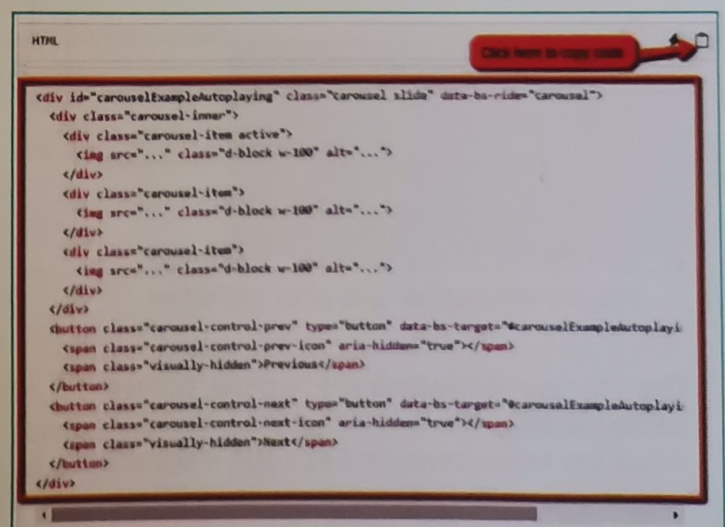
- b. So, within the Carousel window, scroll further down until you see the 'Autoplaying carousels' section.

- c. Within this section, you will find the HTML code for creating the Carousel. Copy the code by clicking on the clipboard looking button.



Autoplaying carousels

You can make your carousels autoplay on page load by setting the `ride` option to `carousel`. Autoplaying carousels automatically pause while hovered with the mouse. This behavior can be controlled with the `pause` option. In browsers that support the [Page Visibility API](#), the carousel will stop cycling when the webpage is not visible to the user (such as when the browser tab is inactive, or when the browser window is minimized).



Activity

e. Similarly, we will also copy the JavaScript link, provided to us by Bootstrap.

CDN links

As reference, here are our primary CDN links.

Description	URL
CSS	https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css
JS	https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/js/bootstrap.bundle.min.js

f. To add the JavaScript link, create a `<script>` tag just before the closing `</body>` tag and paste the link as value for the 'src' attribute.

```
<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/js/bootstrap.bundle.min.js"></script>
</body>
</html>
```

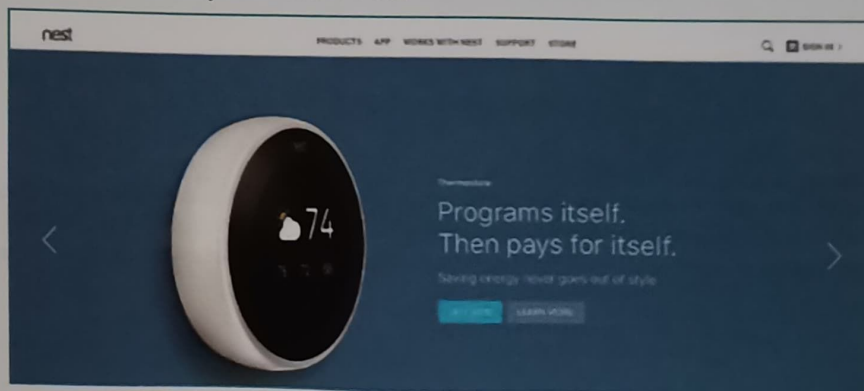
These 2 links contain pre-built Bootstrap components. We must include them on an html file. Without them, we can't use Bootstrap features on a website.

Now, we can go ahead and start using Bootstrap functionalities.

3 Add an image carousel using Bootstrap

What is an image carousel?

An image carousel is a slideshow that cycles through several images, displaying one image at a time. We can set images to transition automatically or manually with button clicks. An image carousel is a trending design concept that many websites use.

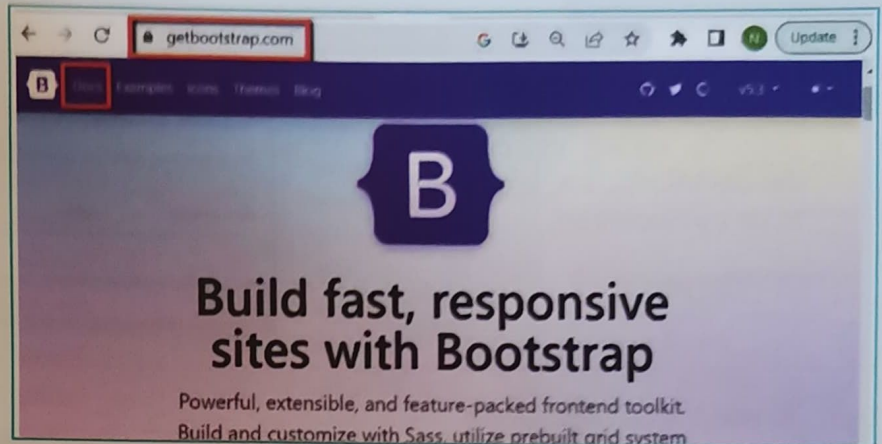


Credits: www.rit.edu

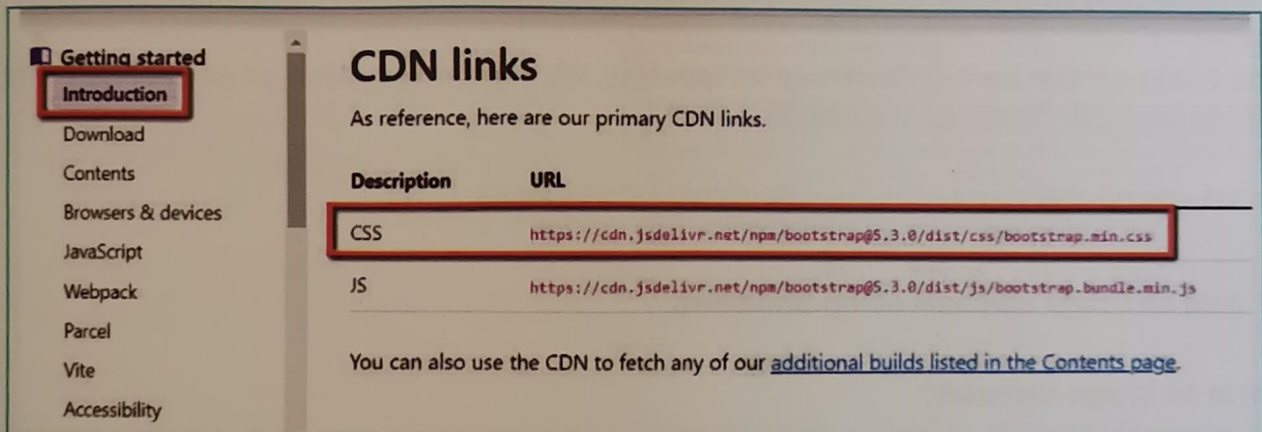
We can use HTML and CSS to create an image carousel. But in this lesson, we will use Bootstrap since it makes the process simpler.

Activity

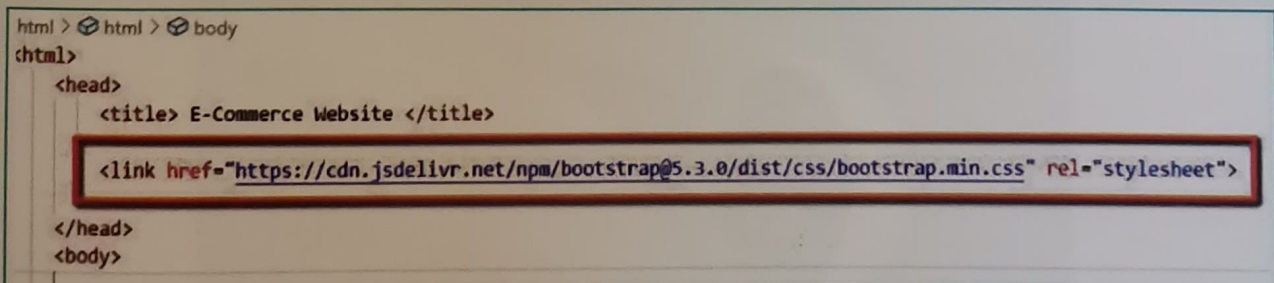
- b. Open the first link. This redirects to Bootstrap's official website '<https://getbootstrap.com>'. Here, click on 'Docs', available on the Navigation Bar, which will land you on a new page, referred to as the 'Bootstrap's Documentation Page'.



- c. Within the 'Introduction' menu item present on the Sidebar, scroll down to the 'CDN Links' section and copy the CSS link.



- d. Now, in the index.html page, create a <link> tag within the <head> tag and paste the copied link as value for the 'href' attribute.



Activity

- b. Next, create a div container with a class "main". This will be the main container, with all our other content placed inside it.

```
<html>
<head>
  <title> E-Commerce Website </title>
</head>
<body>

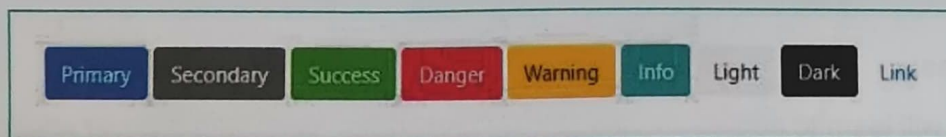
  <div class="main">

  </div>
```

2 Using Bootstrap

Bootstrap's website defines it as, "The most popular HTML, CSS, and JS framework for developing responsive, mobile-first projects on the web".

Bootstrap is a huge collection of pre-built HTML and CSS components, readily available to use for personal websites. HTML components like buttons, lists, tables, forms, etc are available to use. The components on bootstrap are responsive, which means that their size and position on a webpage automatically changes on small screen devices like smartphones and tablets.



Bootstrap Buttons

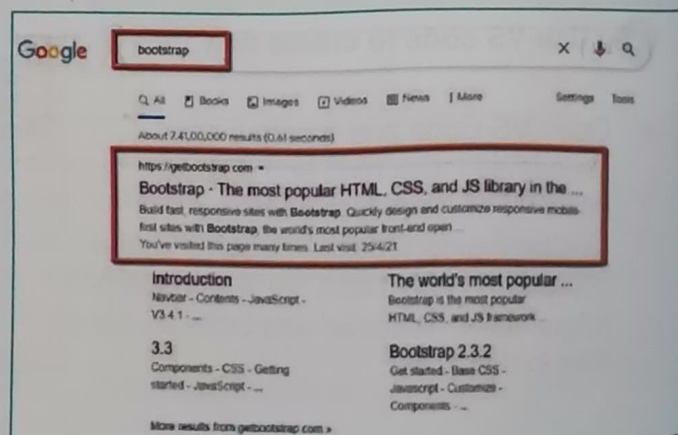
<https://getbootstrap.com/docs/5.3/components/buttons/#base-class>

Visit <https://getbootstrap.com/> to view all the available components.

Now, let us integrate Bootstrap on our website.

To use Bootstrap features, we must first integrate it on our website. The simplest way to do so, is by including the provided **CSS** and **JavaScript** links available on Bootstrap's website.

- a. Open a browser and search for **Bootstrap**.



Examples of static websites:



The fixed appearance of a site may become boring for a user who is visiting it multiple times. It would be helpful if we could implement some basic, non-static functionalities such as an image slider or carousel, contact or register forms that takes user input and stores it in a server, pop-up boxes, simple animations/effects and so on. These features tend to give the site a dynamic appearance and aid in retaining a visitor's attention.

Dynamic Features:

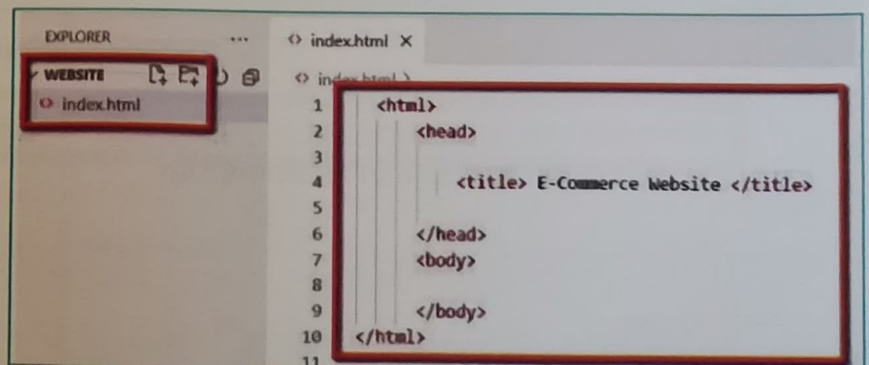
In this lesson, we will learn to add an image carousel and create cards with a hover effect.

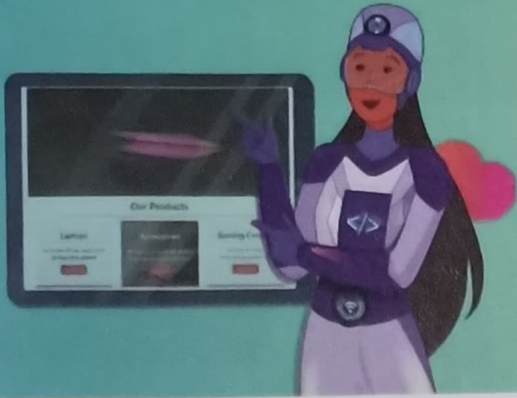
Activity

Create a website for a computer store

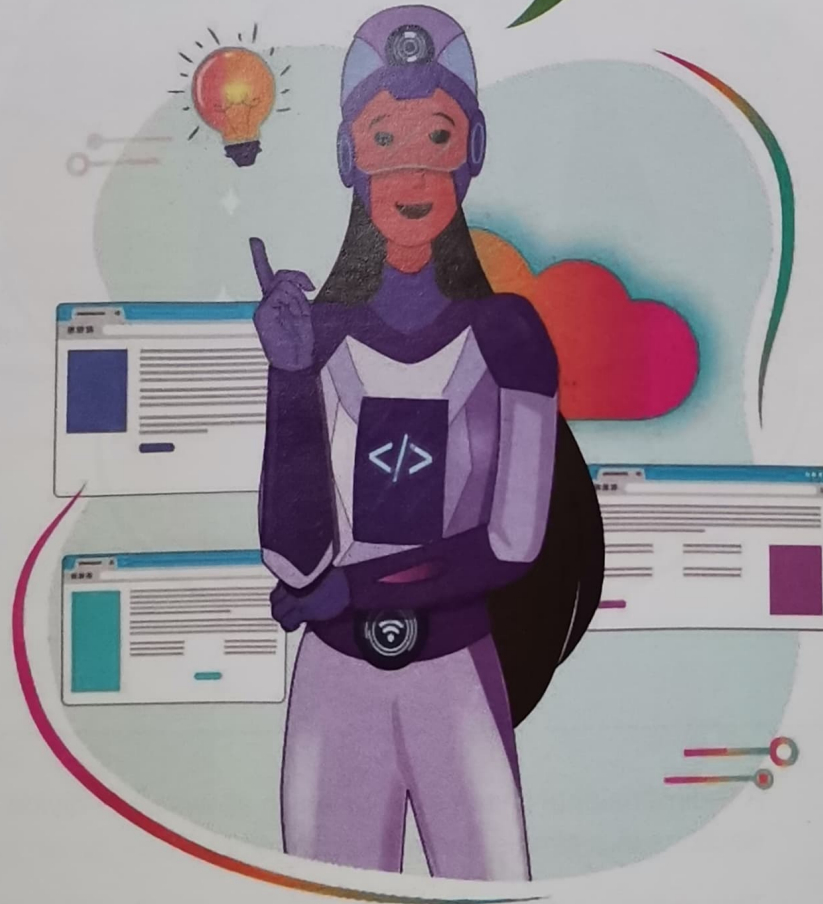
1 Use VS code to create new files

- Open VS Code and create a new file named `index.html` and add the basic HTML tags along with the title "E-Commerce Website". Save it on the computer. In our case, we will create a folder called 'Website' and store all the files in that folder.





We can add dynamic components to a webpage to make it look more user friendly, interactive, and intuitive. These elements ensure that the website does not look dull.



A static webpage is a web page that is delivered to the user exactly the way it is stored. It is also called a flat page or stationary page. A static webpage is created using HTML (for the structure) and CSS (for the design). Individuals and small business houses use static websites to publish information about their companies and products.